

Alops - Module 1 Assignment

Question 1:

- (1) (a) Entanglement (CACE) - because changing one feature changed everything
(b) Undeclared Consumers - because Other teams are consuming our model silently
(c) Configuration & Glue-Code Debt - Many undocumented scripts create a tangled pipeline
- (2) I choose the third one .And to mitigate this we can put all the training scripts and configuration under git version control and create a conda/mamba environment that specifies the required environment. The pipeline steps and configurations should be committed together so the exact code used for training can be reproduced

Question 2:

(2) Among the 6 experiments the best performing run is the one that used a learning rate of 0.001 and batch size 64, achieving a validation accuracy of 0.9747 and macro F1 score of 0.9745. The other runs with learning rate 0.001 also performed in a similar fashion, while increasing learning rate to 0.01 resulted in lower validation accuracy for all three batch sizes. From this we can infer that smaller learning rate allowed MLP to converge more effectively for this dataset .

The training-loss curves show that training loss decreases throughout training for all runs, with the 0.001 learning-rate models generally reaching lower final losses. Validation accuracy initially improves and then mostly plateaus with some fluctuations. Since validation accuracy does not show a strict decrease while training loss continues decreasing, there is no strong evidence of severe overfitting.

There is, however, some sign of diminishing validation improvement in later iterations.

Comparing hyperparameters, learning rate appears to have the larger effect on performance. For example, with batch size 32, reducing the learning rate from 0.01 to 0.001 improves test accuracy from approximately 0.9619 to 0.9742. In comparison, changing batch size at learning rate 0.001 changes accuracy only slightly, from 0.9731 to 0.9747. Therefore, learning rate was the more influential hyperparameter in these experiments.

- (3)

```
mlflow.log_param("learning_rate", learning_rate)
mlflow.log_param("batch_size", batch_size)
mlflow.log_param("hidden_layer_sizes", hidden_layers_sizes)
mlflow.log_param("max_iter", max_iter)
mlflow.log_metric("accuracy", acc)
mlflow.log_metric("f1_macro", f1)
mlflow.log_metric("train_loss", model.loss_curve_[-1])
mlflow.log_metric("val_accuracy_final", model.validation_scores_[-1])
for step, loss in enumerate(model.loss_curve_):
    mlflow.log_metric("train_loss_curve", loss, step=step)
for step, val_acc in enumerate(model.validation_scores_):
    mlflow.log_metric("val_accuracy", val_acc, step=step)
```

Question 3:

Partner B reproduced the experiment and obtained metric values that were very close to Partner A's results. The differences were only due to **display/rounding precision**, with Partner A's reported values being truncated versions of the values obtained by Partner B. Therefore, the results were considered a **match within the stated tolerance**, confirming that the experiment was reproducible.